MetaFeature-Orchestrator

Automated Evaluation and Agentic Prompt Orchestration for Large-Scale AI

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Executive Summary

Generative AI capabilities are now embedded across Apple's product ecosystem at unprecedented scale. Hundreds of AI-powered features—spanning writing assistance, visual generation, translation, summarization, and contextual automation—are deployed globally across dozens of languages and cultural contexts. These systems evolve rapidly and exhibit increasing levels of autonomy. Under these conditions, traditional evaluation approaches—reliant on manually authored prompts, ad hoc human review, or loosely governed automated judges—no longer satisfy Apple's standards for quality, safety, or governance.

This report introduces the MetaFeature-Orchestrator, a metadata-driven evaluation infrastructure that automates the generation of evaluation prompts and enforces Responsible AI (RAI) requirements by design. The system treats evaluation prompts as governed, versioned contracts, derived systematically from explicit feature metadata, canonical feature taxonomies, and a centralized metric registry. By encoding human evaluation standards into a deterministic orchestration layer, the MetaFeature-Orchestrator guarantees auditability, reproducibility, and cross-feature consistency. Critical RAI constraints—including hallucination detection, privacy protection, and bias assessment—are injected structurally and cannot be bypassed by individual feature teams.

By transforming evaluation from an ad hoc activity into durable platform infrastructure, the MetaFeature-Orchestrator enables generative AI innovation at scale without compromising quality, user trust, or ethical commitments. Designed to evolve alongside increasingly agentic AI systems, the framework provides a future-proof foundation for trajectory-level evaluation and autonomy assessment as intelligent capabilities continue to expand across the ecosystem.

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1. Problem Statement and Strategic Context

1.1 The Limits of Manual and Ad-Hoc Evaluation

Early generative AI systems were evaluated using manually written prompts, qualitative human review, and feature-specific heuristics. While effective for experimentation or isolated use cases, these approaches do not scale to Apple's current deployment footprint. Today's environment is characterized by:

- Scale: Hundreds of AI-enabled features across multiple product surfaces.
- Diversity: Dozens of languages, regional norms, and cultural expectations.
- Velocity: Rapid model and feature iteration, often on weekly or monthly cadences.
- Risk: Increased potential for user harm due to hallucination, privacy leakage, bias, or inappropriate tone.

At this scale, evaluation can no longer be treated as downstream quality assurance. It must function as core infrastructure that ensures consistency, comparability, and accountability across the entire ecosystem.

1.2 The Limits of LLM-as-Judge Without Prompt Governance

Automated evaluation using large language models as judges addresses reviewer scalability but does not solve the underlying governance problem. LLM-based judges are highly sensitive to prompt phrasing, omitted constraints, and ambiguous scoring instructions. Without systematic control over prompt generation:

- Evaluation criteria drift over time.
- Safety and privacy checks are silently omitted.
- Scores cannot be reliably compared across teams or releases.

The core challenge is therefore not a shortage of evaluators, human or automated, but the absence of a governed evaluation layer that makes quality standards explicit and enforceable.

2. Feature Taxonomy and Risk Model

2.1 Why Taxonomy Is Foundational

Scalable evaluation requires a precise and shared definition of *what quality means* for different classes of generative behavior. In large-scale generative AI systems, quality is not a universal concept—it is task-dependent, risk-dependent, and context-dependent. A single, generic evaluation rubric cannot adequately capture the distinct objectives and failure modes of summarization, translation, creative generation, or autonomous reasoning.

For this reason, the MetaFeature-Orchestrator treats feature taxonomy as a governing abstraction, not a descriptive label applied after the fact. Taxonomy determines *how* a feature is evaluated, *which risks are considered non-negotiable*, and *which metrics are valid*. Without a well-defined taxonomy, evaluation systems inevitably drift toward lowest-common-denominator metrics that obscure real failures and provide false confidence.

Within the Orchestrator, each feature category explicitly defines:

- Applicable quality metrics aligned with the feature's behavioral intent
- Dominant failure modes that must be surfaced and penalized
- Mandatory Responsible AI constraints, including safety, privacy, and bias checks
- Evaluation prompt structure, ensuring consistent judgment semantics

This structure ensures that features with fundamentally different objectives are never evaluated using incompatible standards, even if they share similar interfaces or underlying models.

From an operational perspective, quality metrics must be stable but not static. Metrics should evolve deliberately over time as product understanding deepens, failure modes are discovered, and organizational standards mature. Uncontrolled metric churn undermines longitudinal analysis, while frozen metrics fail to capture emerging risks. The Orchestrator therefore enforces metric versioning and governance, enabling controlled evolution without breaking comparability.

Metric definition follows a human-in-the-loop governance model. Feature owners propose candidate metrics based on deep domain and product knowledge. These proposals are reviewed and validated by a centralized evaluation team to ensure cross-feature consistency

and alignment with platform standards. As system complexity grows, AI assistance can be used to *suggest* metrics or surface overlooked risk dimensions; however, final metric approval remains a human responsibility. This preserves accountability while allowing AI to augment, rather than replace, expert judgment.
By elevating taxonomy to a first-class control mechanism, the MetaFeature-Orchestrator ensures that evaluation remains semantically meaningful, risk-aware, and governable at scale—despite the rapid growth and diversification of generative AI capabilities.

2.2 Apple's AI Features

Apple Intelligence represents a system-wide integration of generative and assistive AI capabilities across iPhone, iPad, Mac, Apple Watch, and Vision Pro. These features span communication, creativity, productivity, and contextual automation, and are deeply embedded into core operating system workflows rather than delivered as standalone applications. Apple's approach emphasizes on-device processing, privacy preservation, and progressive global rollout across languages and regions.
Core Capability Domains

- **Language Understanding and Communication**: Apple has introduced AI-powered features that enhance everyday communication while maintaining privacy and user control. These include real-time Live Translation across Messages, FaceTime, and phone calls; Smart Reply suggestions in Mail and Messages; and auto-generated summaries for emails, message threads, voicemails, notes, and call recordings. Collectively, these features focus on semantic fidelity, tone appropriateness, and time-saving assistance rather than free-form generation.
- **Writing and Reading Assistance**: System-wide Intelligent Writing Tools allow users to summarize, rewrite, proofread, and restructure text in different tones directly within any text field, including third-party apps. These tools support tasks such as extracting key points, improving clarity, or adjusting formality, and are designed as assistive transformations rather than autonomous content creation.
- **Visual Intelligence and Image Generation**: Apple has expanded AI capabilities into visual understanding and creation. Visual Intelligence enables users to act on what they see—extracting text from images, translating on-screen content, creating calendar events from posters, or querying on-screen context. Generative features such as Image Playground, Genmoji, and Image Wand allow users to create images, emojis, and illustrations from prompts or sketches, with strong emphasis on safety, style control, and contextual relevance.
- **Photo and Media Intelligence:** In Photos, Apple Intelligence supports semantic search, object removal (Clean Up), and auto-generated Memory movies that assemble photos and videos into narrative summaries based on user prompts. These features combine generative reasoning with personal data understanding, requiring careful evaluation of correctness, relevance, and privacy boundaries.
- **Contextual Assistance and Automation:** Apple is evolving Siri and system automation toward deeper contextual awareness. Enhancements include typed Siri input, device how-to assistance, on-screen context understanding, and integration

with personal content such as messages, addresses, and schedules. Intelligent Shortcuts allow users to embed AI actions—such as summarization or image generation—directly into multi-step workflows, enabling increasingly agentic behavior across apps.

- **Health, Fitness, and Personal Coaching**
  Features such as Workout Buddy demonstrate Apple's use of real-time, personalized AI in sensitive domains. These capabilities analyze live user data to generate adaptive coaching and feedback, highlighting the need for robust evaluation around correctness, appropriateness, and user trust.
- **Developer and Platform Enablement**
  Apple has opened access to its on-device foundation models through APIs, enabling third-party developers to build AI-powered features that run locally and preserve privacy. This expands the AI surface area beyond Apple-owned features and increases the importance of consistent evaluation standards across first- and third-party use cases.

Global Rollout and Privacy Model

Apple Intelligence is being rolled out in phases across languages and regions, with initial support for major languages and gradual expansion toward near-global coverage. Not all features are immediately available in all regions, reflecting regulatory, linguistic, and infrastructure considerations. A defining characteristic of Apple's AI strategy is its privacy-first architecture: most inference runs on-device using Apple silicon, with heavier workloads handled through Private Cloud Compute, where data is processed transiently and not retained. This hybrid model introduces unique evaluation challenges around consistency, determinism, and safety across heterogeneous execution environments.

2.3 Canonical Feature Categories

Apple's AI feature portfolio spans a wide range of generative and assistive behaviors—including summarization, translation, content generation, reasoning, and automation—each with distinct objectives, failure modes, and risk profiles. As these capabilities are deployed globally and exhibit increasing autonomy, evaluation cannot rely on a single, uniform standard. Instead, it requires a taxonomy-driven framework that explicitly defines what quality means for each class of behavior.

The MetaFeature-Orchestrator operationalizes this principle by organizing features into a set of canonical categories. Each category serves as a governing abstraction that determines the applicable evaluation metrics, dominant risks, and required Responsible AI constraints. This ensures that features with fundamentally different intents are never evaluated using incompatible or overly generic criteria.

Category	Representative Features	Primary Evaluation Focus
Summarization	Email summaries, article digests	Faithfulness, coverage, hallucination avoidance
Auto-Reply & Writing Assistance	Smart Reply, Writing Tools	Relevance, tone, privacy, social appropriateness

Translation	Live translation, message translation	Semantic equivalence, fluency, cultural alignment
Assistant & Reasoning	Task planning, shortcuts	Correctness, grounding, reasoning validity
Creative Generation	Image Playground, Genmoji	Safety, bias, policy compliance

A controlled custom category is provided for novel or emerging feature types. Features assigned to this category must explicitly declare their evaluation metrics and are subject to mandatory Responsible AI enforcement, ensuring that innovation does not bypass governance or safety standards.

3. System Architecture Overview

3.1 Design Principles

The MetaFeature-Orchestrator is designed as **evaluation infrastructure**, not an experimental agent. Its architecture is governed by a small set of non-negotiable principles that ensure reliability, scalability, and long-term governance:

- **Statelessness**
  The orchestration layer maintains no hidden or implicit memory. All state—feature definitions, metric configurations, prompt versions, and evaluation results—is persisted in externally governed stores. This simplifies reasoning about system behavior and enables horizontal scalability.
- **Auditability and Traceability**
  Every evaluation outcome is fully traceable to the exact prompt template, metric definitions, and version identifiers used to produce it. This guarantees reproducibility, supports regression analysis, and enables compliance review.
- **Structural Responsible AI Enforcement**
  Safety, privacy, and bias constraints are injected by construction rather than configured ad hoc. Responsible AI requirements are enforced uniformly across features and cannot be bypassed at the feature level.
- **Cost Awareness**
  The system is optimized to minimize unnecessary model invocation. Prompt generation is performed offline and deterministically wherever possible, reserving model calls for evaluation execution rather than orchestration logic.

Together, these principles ensure that evaluation remains consistent, defensible, and economically viable as the number of features and supported locales grows.

3.2 Dual-Mode Architecture

To balance determinism with extensibility, the MetaFeature-Orchestrator operates in two complementary modes that share a common execution pipeline and output schema:

- **Template Mode**
  A deterministic prompt writer generates canonical evaluation contracts from

predefined, versioned templates. This mode is optimized for standard feature categories and provides maximum reproducibility, consistency, and operational efficiency.

- **Agentic Mode**
  A controlled reasoning agent is used for novel or complex features that cannot be fully resolved through static templates. The agent selects metrics, constraints, and templates through explicit, auditable tool invocation. Importantly, this mode remains deterministic at the prompt-generation layer, preserving reproducibility despite dynamic reasoning.

Both modes produce structurally identical evaluation prompts and feed into the same downstream evaluation and scoring pipeline. This separation allows the system to accommodate innovation and feature diversity without compromising governance or consistency.

4. Prompt Orchestration Methodology

4.1 Feature Metadata Schema

Each AI feature is onboarded through a structured metadata descriptor that serves as the **authoritative input** to the evaluation system. Rather than relying on manually crafted prompts, the MetaFeature-Orchestrator derives evaluation behavior directly from this metadata, ensuring consistency and governance at scale.

The metadata schema captures four essential dimensions:

- **Feature Domain**
  Identifies the canonical feature category and determines the applicable metric families and evaluation semantics.
- **Behavioral Intent**
  Explicitly defines the feature's success criteria, clarifying what constitutes correct or high-quality behavior.
- **Locale Context**
  Encodes language- and culture-specific constraints, enabling evaluation to account for regional norms, linguistic formality, and localized expectations.
- **Responsible AI (RAI) Constraints**
  Declares prohibited behaviors, elevated risk flags, and domain-specific safety requirements.

This metadata functions as the **single source of truth** for evaluation generation, eliminating ambiguity and preventing feature-specific divergence in evaluation standards.

4.2 Template Grammar

All evaluation prompts generated by the Orchestrator adhere to a strict, shared grammar. This grammar defines not only the structure of the prompt, but also the semantics of judgment, ensuring that evaluation results are reproducible and comparable across features and releases.

Each prompt is composed of the following elements:

- **Judge role and authority definition**
- **Grounding rules and explicit prohibitions** (e.g., no external knowledge)
- **Feature and locale context**
- **Metric definitions with explicit scoring instructions**
- **Standardized scoring scale**
- **Strict JSON output schema**
- **Evaluation inputs** (source content and model output)

Flexibility is intentionally constrained at this layer. Allowing ad hoc prompt variation introduces uncontrolled variance, undermines longitudinal analysis, and re-introduces evaluator drift. Rigid prompt grammar is therefore a prerequisite for reliable large-scale evaluation.

4.3 Responsible AI Injection

Responsible AI requirements—including privacy protection, bias mitigation, hallucination detection, and safety enforcement—are injected automatically based on the feature's category and declared risk profile. These constraints are applied structurally during prompt generation and **cannot be disabled or overridden** by individual feature teams.
By embedding Responsible AI enforcement directly into the orchestration layer, the system ensures uniform application of ethical and safety standards across the entire AI ecosystem.

5. Advanced Evaluation Techniques

5.1 Hallucination Detection via Claim Extraction

For content-preserving tasks such as summarization and translation, the Orchestrator employs a claim-level evaluation strategy. Model outputs are decomposed into atomic factual claims, each of which is evaluated against the source context to identify unsupported, fabricated, or contradictory information. This approach enables fine-grained hallucination scoring and avoids the blind spots of holistic similarity metrics.

5.2 Cross-Lingual Evaluation

To support global deployment, the system evaluates non-English outputs using cross-lingual semantic representations rather than relying on English back-translation. This design reduces English-centric bias, preserves local linguistic nuance, and enables more faithful assessment of meaning across languages.

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6. Operational Workflow

The MetaFeature-Orchestrator operates as a closed-loop evaluation pipeline designed to support continuous quality assurance and long-term governance at scale:

- **Onboarding**
  Feature teams submit structured metadata descriptors that define feature intent, category, locale context, and Responsible AI constraints.

- **Prompt Synthesis**
  The Orchestrator generates governed, versioned evaluation prompts deterministically based on the submitted metadata and canonical templates.
- **Execution**
  Evaluation prompts are executed against model outputs to produce structured, machine-readable scores and diagnostics.
- **Validation**
  Targeted human review samples are used to calibrate automated judges and verify alignment with human quality standards.
- **Governance**
  Evaluation results, prompt versions, and metric configurations are persistently stored, versioned, and audited.

This closed-loop workflow enables reliable regression tracking, trend analysis across releases, and continuous monitoring of quality and risk over time.

7. Governance and Ownership

Evaluation standards within the MetaFeature-Orchestrator are centrally governed to ensure consistency and defensibility across the ecosystem. Metric definitions, prompt templates, and Responsible AI constraints are owned, reviewed, and versioned by a centralized evaluation authority.

Feature teams retain responsibility for providing accurate and complete metadata that reflects their feature's behavior and intent. However, they cannot override or bypass evaluation standards or Responsible AI requirements. This clear separation of responsibilities preserves product-level flexibility while maintaining organizational consistency, comparability, and accountability.

8. Conclusion and Roadmap

The MetaFeature-Orchestrator transforms generative AI evaluation from an ad hoc, feature-specific activity into durable platform infrastructure aligned with Apple's quality bar and ethical commitments. By encoding human judgment into governed, deterministic systems, it enables innovation at scale without compromising safety, trust, or accountability.

Looking forward, the framework is designed to evolve alongside increasingly agentic AI capabilities. Planned extensions include real-time evaluation integrated into experimentation pipelines, automated adversarial testing to surface rare but high-impact failure modes, and trajectory-level assessment for autonomous, multi-step behaviors. Together, these capabilities establish a future-proof foundation for the next generation of intelligent computing.

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